

CRYSTAL AM TRANSMITTER

■ PRADEEP G.

Here is the circuit of a medium-power AM transmitter that delivers 100-150 mW of radio frequency (RF) power.

At the heart of the circuit is a crystal oscillator. A 10MHz crystal is used to generate highly stable carrier frequency. Audio signal from the condenser mic is amplified by the amplifier built around transistors T1, T2 and T3. The amplified audio sig-

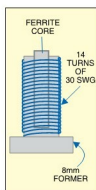


Fig. 2: Oscillator coil

nal modulates the RF carrier generated by the crystal oscillator built around transistor T4. Here modula-

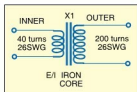


Fig. 3: Modulation transformer

tion is done via the power supply line. The amplitude-modulated (AM) signal is obtained at the collector of oscillator transistor T4.

By using matching dipole antenna and co-axial cable, the range of signal transmission can be increased. For maximum range, use a sensitive radio with external wire antenna.

The circuit works off a 9V-12V battery. For oscillator coil L1, wind 14 turns of 30SWG wire round an 8mm diameter radio oscillator coil former with a ferrite bead (see Fig. 2). For modulation transformer X1, you can use the audio output transformer of your old transistor radio set. Alternatively, you can make it from E/I section transformer lamination with inner winding having 40 turns of 26SWG wire and the outer winding having 200 turns of 30SWG as shown in Fig 3.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. ●

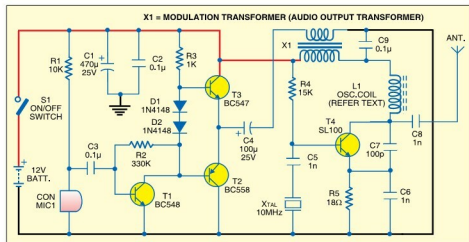


Fig. 1: Circuit of crystal AM transmitter

KEEP AWAY Ni-Cd FROM MEMORY EFFECT

■ T.A. BABU

Ni-Cd batteries suffer from an undesirable memory effect due to partial discharge. The remedy is to completely discharge the battery before recharging.

A simple resistor will do the discharging job nicely but there is the risk that the cell may discharge beyond a certain voltage and cause polarity reversal in the cell. Here is a simple circuit that discharges the cell to a level of 600 mV, ensuring that the cell

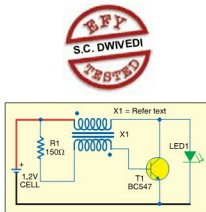


Fig. 1: Circuit to keep Ni-Cd away from memory effect

is correctly discharged without the risk of polarity reversal.



Fig. 2: Winding details

The battery is not discharged at a constant current. This allows the battery to recover during the intervals, to extend its